Ethan Debeb

https://debeb01.github.io | http://www.linkedin.com/in/ethan-d-715631316

EDUCATION

First Year Engineering, **Purdue University,** West Lafayette, IN
BS Aeronautics and Astronautics (Intended), *Junior academic standing* May 2030

The Center for Advanced Studies in Science, Math, & Technology at Wheeler High School Marietta, GA
GPA: 3.443/4.088 | ACT: 36 | May 2026
Relevant Courses: AP Calculus AB & BC, AP Physics C: Mechanics, AP Computer Science, Advanced Physics Robotics

RESEARCH & PUBLICATIONS

A. Khalid, O. Ahmed, and **E. Debeb**, "Real-Time Vehicle Detection and Enumeration Using Unmanned Aerial Systems," presented at the **AIAA AVIATION** 2026 Forum, San Diego, CA, June 2026. DOI: https://doi.org/10.2514/6.2026-4753

EXPERIENCE

Aerospace Research Intern Jul 2025 – Dec 2025
AERO Lab, Kennesaw State University Marietta, GA

- Designed and evaluated a multirotor UAV using SolidWorks, applying computational fluid dynamics (CFD) to analyze lift, drag, and aerodynamic forces and predict aircraft performance prior to build,
- Integrated and evaluated YOLO object-detection models on UAV platforms to identify and count roadway vehicles, supporting computer vision research for traffic monitoring and safety applications,
- Designed a UAV life jacket delivery mechanism in SolidWorks and performed structural stress analysis to verify that the commissioned payload remained within the mechanism's load carrying capacity,
- Contributed to the design, build, and testing of a gravity powered pumpkin launcher for the 15th Annual Pumpkin Launch competition.

Quality Engineer Shadow Jun 2024
Porsche Cars North America Atlanta, GA

- Shadowed QE interns, assisted with vehicle issue identification, diagnostic data review and root cause analysis

LEADERSHIP AND MEMBERSHIP

Purdue Space Program (PSP), IREC Team West Lafayette, IN
Structures Sub-Team Member August 2026 - Present

- Assist with researching and designing a structural component and layout brackets for a competitive rocket targeting a 10,000/30,000 ft apogee

Students In Space Marietta, GA
President (SY 25-26), **Vice President** (SY 24-25)

- Led the Students in Space club as president, directed planning initiatives while mentoring new members and serving as the primary liaison between the team and club leadership.
- Directed project design for aerospace competitions, organized meetings, and coordinated cross-team efforts from concept through build and final presentation.
- Founded and managed the Students in Space Podcast, overseeing content and production.

Competed in the 2025 Canada Space Design Competition and 2024 Georgia Tech/NASA Payload Challenge

CircuitRunners Robotics, Inc, FIRST Tech Challenge -1002 Surge Marietta, GA
Lead Designer & CAD Builder Aug 2022 – May 2026

- Directed robot design and mechanism development to 14 members as a Lead Designer, using SolidWorks for CAD modeling, Python for robot programming, and printing for rapid prototyping – guiding the team to three F**IRST World Championship qualifications (2023, 2024, 2026) and three state championships.**
- Mentored 2-4 underclassmen in CAD design, build accuracy, and troubleshooting, provided training for new members to integrate hardware builds with software systems.
- Directed marketing strategy to promote the team and robotics programs to external audiences, contributing to award-winning seasons.

PROFESSIONAL AFFILIATIONS

Member, **National Society of Black Engineers (NSBE)** – Purdue University Chapter, *2022 - Present*
Member, **American Institute of Aeronautics and Astronautics (AIAA)** – Purdue University Chapter, *2025 - Present*

TECHNICAL SKILLS

Programming: Python, Java, C++
Software: SolidWorks, CFD, structural stress analysis OnShape CAD, Visual
Tools: Studio Code, Google Suite